more explain the fundamental particle properties of Tables 1.1 and 1.2 than the Dow Jones average today can explain your initial investment in stocks ten years ago.

In fact, had experiments revealed a somewhat different particle content in the microscopic world, possibly interacting with somewhat different forces, these changes could have been fairly easily incorporated in the standard model by providing the theory with different input parameters. The structure of the standard model, in this sense, is too flexible to be able to explain the properties of the elementary particles, as it could have accommodated a range of possibilities.

String theory is dramatically different. It is a unique and inflexible theoretical edifice. It requires no input beyond a single number, described below, that sets the benchmark scale for measurements. All properties of the microworld are within the realm of its explanatory power. To understand this, let's first think about more familiar strings, such as those on a violin. Each such string can undergo a huge variety (in fact, infinite in number) of different vibrational patterns known as resonances, such as those shown in Figure 6.1. These are the wave patterns whose peaks and troughs are evenly spaced and fit perfectly between the string's two fixed endpoints. Our ears sense these different resonant vibrational patterns as different musical notes. The strings in string theory have similar properties. There are resonant vibrational patterns that the string can support by virtue of their evenly spaced peaks and troughs exactly fitting along its spatial extent. Some examples are given in Figure 6.2. Here's the central fact: Just as the different vibrational patterns of a violin string give rise to different musical notes, the different vibrational patterns of a fundamental string give rise to different masses and force charges. As this is a crucial point, let's say it again. According to string theory, the properties of an elementary "particle"—its mass and its various force charges—are determined by the precise resonant pattern of vibration that its internal string executes.

It's easiest to understand this association for a particle's mass. The energy of a particular vibrational string pattern depends on its amplitude—the maximum displacement between peaks and troughs—and its wavelength—the separation between one peak and the next.